

MAI MOHAMMED

AI Engineer | Machine Learning Engineer | NLP & Generative AI Specialist

Cairo, Egypt • +20 106 576 1643 • maimohamed999999@gmail.com • linkedin.com/in/mai-mo7med • github.com/MaiM0hamed

PROFESSIONAL SUMMARY

Results-driven **AI Engineer** with production experience in **NLP**, **Large Language Models (LLMs)**, **Retrieval-Augmented Generation (RAG)**, **Computer Vision**, and **Generative AI**. Demonstrated ability to reduce model latency by **40%**, achieve **94% detection accuracy**, and deliver **95% task-completion rates** in live voice AI systems. Secured **EGP 30,000 ITIDA funding** and ranked **4th at the Dell Middle East AI Competition** across 200+ teams. Proficient across the full AI/ML lifecycle — data preprocessing, model fine-tuning, vector database integration, MLOps, cloud deployment, and CI/CD — with deep expertise in **Python**, **PyTorch**, **TensorFlow**, **LangChain**, **Hugging Face**, and **OpenAI APIs**. Passionate about building scalable, production-ready AI solutions that drive measurable business value.

CORE AI COMPETENCIES

Generative AI & LLMs: LLM Fine-tuning (LoRA/QLoRA), RAG Systems, Prompt Engineering, Multi-Agent Orchestration (LangGraph, AutoGen), OpenAI API, Gemini API

Computer Vision: Object Detection (YOLOv8/v11), Segmentation (Mask2Former), Classification (ResNet50), OCR (Tesseract), FaceNet, Model Pruning & INT8 Quantization

Vector DBs & Search: FAISS, ChromaDB, Semantic Search, Sentence Transformers, Hybrid Retrieval, Advanced Chunking, Embedding Caching, Namespace Isolation

Data Science: Python, Pandas, NumPy, Scikit-learn, SQL, Statistical EDA, Matplotlib, Seaborn, Power BI, Tableau, Feature Engineering, Web Scraping

NLP & Speech AI: Transformers, BERT/GPT Fine-tuning, Sentiment Analysis, NER, Intent Recognition, Whisper STT, gTTS, Coqui TTS, LangChain Pipelines

ML Engineering: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, MLflow, CNN/RNN/LSTM/GAN, Time Series Forecasting, Hyperparameter Tuning

MLOps & Cloud: Docker, CI/CD, Azure AI Services, Twilio Cloud Telephony, MLflow Tracking, REST APIs, Flask, Streamlit, Git, Model Deployment & Scaling

Additional Stack: C++, C, MATLAB, Clean Architecture, Agile/Scrum, Problem Solving (150+ DSA), Technical Curriculum Design, API Integration

PROFESSIONAL EXPERIENCE

AI Engineer | VO Technology

Feb 2026 – Present • Cairo, Egypt

ERP AI Integration • Python • Deep Learning • Predictive Analytics • Data Pipelines • Azure AI

- Architected and deployed AI/ML components within enterprise ERP systems, automating complex workflows and reducing manual processing time across departments.
- Built deep learning predictive modules for **demand forecasting**, **anomaly detection**, and resource optimization, enabling proactive business decision-making.
- Designed scalable data pipelines and optimized database schemas to support high-throughput AI inference and improve system efficiency under peak load.
- Delivered AI-powered real-time reporting dashboards, improving business intelligence accuracy and significantly reducing report generation time.
- Collaborated with cross-functional product and engineering teams in Agile sprints to align AI deliverables with enterprise digital transformation roadmaps.

AI & Data Science Curriculum Designer | Armstrong

Dec 2025 – Apr 2026 • Part-time

Curriculum Design • ML • Deep Learning • NLP • Computer Vision • Generative AI • Python

- Designed end-to-end AI and Data Science programs spanning **Machine Learning**, **Deep Learning**, **NLP**, **Computer Vision**, and **Generative AI** for 100+ learners.
- Built structured learning tracks with curated real-world datasets, hands-on capstone projects covering **RAG system development**, **LLM fine-tuning**, and MLOps deployment.
- Created comprehensive assessment rubrics and project evaluation frameworks, producing measurably stronger AI engineering competency outcomes among graduates.

AI Engineering Intern | ITSOLERA Pvt Ltd

Jun 2025 – Aug 2025

NLP • Time Series Forecasting • Flask • Streamlit • REST APIs • OpenAI • Hugging Face • CI/CD

- Embedded in the Deep Learning team to co-develop **production NLP pipelines** and time-series forecasting models deployed live for real business clients.
- Built and shipped AI microservices (Flask, Streamlit) integrating **OpenAI and Hugging Face APIs** via REST endpoints within CI/CD deployment environments.
- Contributed to model evaluation frameworks — **precision**, **recall**, **F1**, **latency benchmarking** — to validate production readiness, catching 3 critical regressions pre-release.

KEY AI PROJECTS

Retrieval-Augmented Generation (RAG) Chatbot

Generative AI • LLM • Vector Search

OpenAI API • Hugging Face • FAISS • ChromaDB • Sentence Transformers • LangChain • Python

- Engineered a production-grade RAG pipeline grounding **LLM responses in custom knowledge bases**, reducing hallucination rate by **~60%** vs. baseline GPT-4.
- Implemented **hybrid semantic + keyword search** over FAISS and ChromaDB vector stores, achieving **89% retrieval precision** across multi-domain document corpora.
- Applied sliding-window chunking, embedding caching, and async retrieval — reducing average query latency by **42%** under 50+ concurrent users.
- Designed modular **LangChain chains** with plug-and-play LLM backends (GPT-4, Mistral, Llama-3) enabling zero-refactor model swapping across deployment targets.

Smart Assistive System for Visually Impaired

Computer Vision • Multi-modal AI • Edge AI

YOLOv8 • ResNet50 • Mask2Former • Tesseract OCR • FaceNet • gTTS • Coqui TTS • Python

- Developed AI wearable glasses with real-time obstacle detection at **94% accuracy** and **180ms** end-to-end inference latency on CPU-constrained edge hardware.
- Optimized **YOLOv8** inference speed by **44%** via structured pruning and INT8 quantization, enabling real-time performance without GPU acceleration.
- Fused a multi-modal AI pipeline: scene classification (ResNet50), semantic segmentation (Mask2Former), OCR (Tesseract), and face recognition (FaceNet) into one unified system.
- Awarded **EGP 30,000 ITIDA government funding** and **ranked 4th at Dell Middle East AI Competition** (200+ competing teams) for innovation and technical depth.

Syrian Arabic Voice Ordering Agent — Charco Chicken

Conversational AI • Voice AI • NLP

Twilio SIP • Whisper STT • Google Dialogflow • Responsive Voice TTS • Python • Cloud Telephony

- Built a **dialect-aware SIP voice AI agent** for automated restaurant ordering achieving **88% intent accuracy** on Syrian Arabic and **95% successful order completion**.
- Reduced voice response latency from **2.8s → 1.7s (-39%)** through async STT/TTS orchestration, streaming inference, and parallel audio pipeline processing.
- Deployed on Twilio cloud telephony infrastructure, handling **hundreds of concurrent restaurant calls** with zero-downtime uptime throughout peak service hours.

Arabic Customer Support Triage Agent

Agentic AI • NLP • LLMs • FastAPI

Python • FastAPI • PostgreSQL • SQLAlchemy • OpenRouter API • Pydantic • Docker • GitHub Actions

- Developed an **Arabic-language customer support triage agent** for e-commerce using an end-to-end LLM orchestration pipeline covering intent classification, dialect detection, entity extraction, sentiment analysis, urgency assessment, and team routing.
- Built a **tool-calling workflow** integrating knowledge base search, order lookup, automated response drafting, and human escalation mechanisms using OpenRouter LLMs.
- Designed RESTful APIs with **FastAPI** and integrated database operations via **PostgreSQL + SQLAlchemy ORM**; implemented structured logging and JSON trajectory tracking for agent observability and debugging.
- Created evaluation pipelines measuring **precision, recall, F1-score, routing accuracy**, and adversarial robustness; containerized with Docker and automated CI/CD via GitHub Actions.

EDUCATION

B.Eng., Biomedical Engineering | El-Shorouk Academy, Egypt

Cairo, Egypt

Relevant Coursework: Signal Processing, Biomedical Instrumentation, Data Analysis, C/C++ Programming, Mathematics for Engineers, Pattern Recognition, Digital Systems

ACHIEVEMENTS, CERTIFICATIONS & LEADERSHIP

Awards & Recognition

- 4th Place — Dell Middle East AI Competition (200+ teams)
- ITIDA Project Funding — EGP 30,000 (Smart Assistive System for Visually Impaired)
- 72nd Place — ECPC Collegiate Programming Contest Qualifications

Certifications & Training

- Problem Solving & Data Structures and Algorithms
- YOLOv11 Object Detection Development & Deployment
- Shell NXPLORER Innovation & Entrepreneurship Program
- Azure AI Fundamentals — Microsoft

Leadership & Community

- Head of PR & Fundraising — GDSC AI-Azhar University (2024)
- Project Team Leader — Enactus Shorouk Academy (2023)
- Machine Learning Team Member — MSP AI-Azhar (2022)

Languages

Arabic (Native) • English (Professional B2) • Chinese (Professional)